A zero-loss redesign of the upper-index proof

Conditional composition and a small exact support certificate

Research audit in the fork `kolodziejek/berry-esseen-certificates`

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Abstract

A first failure after a sharp finite prefix has zero predecessor decrement. A support-free scalar recurrence then supplies the local score gates without positive-ceiling propagation. We derive a new support certificate, uniform for all actual indices $n \geq 42$, using 14 closed intervals and 672 rational Bernstein coefficients. This removes the finite upper endpoint from the support part of the proposed LOCAL42 route. The resulting strict-tail theorem is conditional on a genuine sharp base and the retained analytic and local numerical implications. It is not an acceptance of the full conjecture.

1 Scope, notation and retained assumptions

For standardized iid summands with finite third absolute moment $\rho = E|X|^3$, write

$$C_n = \sup_P \frac{\sqrt{n}}{\rho(P)} \sup_x \left| P\{(X_1 + \cdots + X_n)/\sqrt{n} \leq x\} - \Phi(x) \right|, \qquad C_{\mathrm{E}} = \frac{3 + \sqrt{10}}{6\sqrt{2\pi}} < \frac{41}{100}.$$

The inequality for C_{E} and the treatment of reflected left limits are recorded in the independent scalar audit. Attainment of an oriented positive closed-CDF maximizing pair when $C_n \geq C_{\mathrm{E}}$ is a separate retained analytic input; this includes equality, not merely strict exceedance.

The proposed uniform LOCAL42 statement is:

$$\begin{gathered} n \geq 42, \quad C_n \geq C_{\mathrm{E}}, \quad 0 \leq n(C_n - C_{\mathrm{E}}) \leq .54, \quad n(C_{n-1} - C_n)_+ = 0, \\ C_n < .4226, \qquad C_{n-1}, C_{n-2} < .4253 \quad \Longrightarrow \quad \text{contradiction.} \end{gathered} \tag{1}$$

Here (1) is conditional on the complete finite 42 resource and comparison predicates and on their analytic licenses, with the new support certificate below replacing the old finite-index support signs. The detailed transfer of the other index-sensitive hypotheses is in `uniform42.md`. It retains weighted resources $nD \leq 42 d_D$, same-law joint gap/noise constraints, both signs, strict antichain widths, mode floor 40, actual strips and moment coefficients, signed normalization repairs, the complete closed TINY/MID/LOW cover, and the zero-noise Bernoulli theorem. None is proved by the scalar recurrence or by a support calculation.

2 The scalar entry and first-failure composition

Lemma 1 (Support-free smoothing)**.** *For every integer $n \geq 2$,*

$$C_n \leq \sqrt{\frac{n}{n-1}} C_{n-1} + \frac{\sqrt{n}}{15(n-1)^{3/2}} + \frac{2\sqrt{n}}{25(n-1)^2}.$$

Proof. Put $v = n - 1$ and $F_v(t) = \Phi(t/\sqrt{v})$. Conditioning on the last summand introduces at most $C_{n-1}\rho/\sqrt{v}$. Taylor's formula through order two and $EX = 0$, $EX^2 = 1$ give

$$\left| EF_v(t - X) - F_v(t) - \tfrac{1}{2} F_v''(t) \right| \leq \frac{\rho}{15 v^{3/2}},$$

because $\|\phi''\|_\infty = \phi(0) < 2/5$. The heat identity $\partial_v F_v = F_v''/2$ gives

$$\left|F_{v+1} - F_v - \tfrac{1}{2}F_v''\right| \le \frac{1}{4\pi v^2} < \frac{2}{25v^2}.$$

Indeed $\|\phi'''\|_\infty \le 2/\pi$ follows from the Gaussian Fourier integral, and the integrated second variance derivative has factor $1/8$. Multiply by $\sqrt{n}/\rho$, use $\rho \ge 1$, and take suprema. The uniform remainder uses only the third absolute moment and holds at every threshold. □

Lemma 2 (Entry at 42)**.** *If $n \ge 42$ and $C_{n-1} \le C_{\mathrm{E}} \le C_n$, then*

$$0 \le n(C_n - C_{\mathrm{E}}) < \frac{3003}{10250} < \frac{3}{10}, \qquad C_n < \frac{73}{175} < .4226.$$

Proof. For $s = \sqrt{n/(n-1)}$, rationalizing $s - 1$ in the preceding lemma yields

$$n(C_n - C_{\mathrm{E}}) \le \frac{n}{n-1}\left(\frac{C_{\mathrm{E}}}{s+1} + \frac{s}{15} + \frac{2s}{25\sqrt{n-1}}\right).$$

Now $s < 81/80$, $\sqrt{n-1} > 6$, $s + 1 > 2$ and $n/(n-1) \le 42/41$. Thus the displayed expression is strictly smaller than

$$\frac{42}{41}\left(\frac{41}{200} + \frac{27}{400} + \frac{27}{2000}\right) = \frac{3003}{10250}.$$

The reserve below $3/10$ is $36/5125$. Also $C_n < .41 + .3/42 = 73/175$, whose reserve below .4226 is $191/35000$. □

Theorem 3 (Conditional strict tail)**.** *Assume $C_j \le C_{\mathrm{E}}$ for $1 \le j \le 41$, and assume the equality-inclusive implication (1) with all of its analytic and numerical premises. Then $C_n < C_{\mathrm{E}}$ for every $n \ge 42$.*

Proof. If the conclusion fails, choose the least integer $n \ge 42$ with $C_n \ge C_{\mathrm{E}}$. Both predecessors are at most C_{E}; when $n = 42$ these are exactly 40 and 41. Hence their caps are strictly below .4253, and $n(C_{n-1} - C_n)_+ = 0$. Lemma 2 supplies stronger excess and current score bounds than (1). The retained equality-inclusive attainment hypothesis and (1) give a contradiction. Choosing the least strict exceedance instead would establish only a weak tail. □

3 Uniform contact polynomials

Let $c = 819/2000 < C_{\mathrm{E}}$ and $N = 42$. The lower inequality follows from $\sqrt{10} > 3081/500 - 3$, $\pi < 22/7$ and $\sqrt{44/7} < 1567/625$: the resulting lower bound for C_{E} exceeds c by $377/3134000$. Let r, b be upper-bound parameters satisfying $\rho \le r$ and $|\beta| \le b$ for the same current law. They are not lower bounds on the actual moments. Put

$$\ell = \frac{6403124237432}{10^{12}}, \qquad S = \frac{202424330939}{200000000000}, \qquad \ell^2 < 41, \quad S^2 > \frac{42}{41}.$$

For $u \ge 5/2$ define

$$A = (u^3 - 3u)/6, \qquad J = u^5 - 15u, \qquad H_4 = u^4 - 6u^2 + 3,$$
$$D_+ = J/(120 \cdot 41), \qquad W = A - \tfrac{15}{2}D_+, \qquad B_2 = H_4^2/(256 \cdot 41),$$
$$E = \frac{51u^6}{14400 \cdot 41\ell} + \frac{51(3u^2 - 2)}{960 \cdot 41\ell} + \frac{u}{8 \cdot 41^2},$$
$$Q = -u^3 + \tfrac{3}{2}ru^2 + 3bu + \tfrac{.384}{42}r.$$

The infinite-upper-index, zero-loss polynomials are

$$G = \left(\tfrac{A}{2} - 3D_+ - \frac{H_4}{12\ell}\right) W + B_2, \tag{2}$$

$$P = \tfrac{2}{5} S^3 (AW + B_2) + S^3 EW + cQW. \tag{3}$$

The retained signed fifth-order contact envelope states that positivity of $A, H_4, J, W, G, -Q, -P$ excludes support at the corresponding u, for both signs and every real raw threshold. Its source is `paper/analytic-supplement.txt`, lines 455–640. The adverse terms are bounded at 42; the favorable fifth-order lower bound is zero for the unbounded index range. Thus (3) has no hidden finite 48 endpoint. The zero predecessor decrement removes the contact-loss term.

For comparison, the old finite polynomial has $D_- = J/(120 \cdot 47)$ and loss $q = .19$. Direct subtraction gives

$$P_{\infty,0} - P_{48,.19} = SW\left(\frac{S^2 J}{4700} - .19r\right), \qquad G_\infty - G_{48} = 3D_- W.$$

This gives an alternative transfer from old finite signs on suitable shorter intervals, proved in `zero-loss-support.md`. The following new certificate proves the infinite polynomials directly and therefore does not need those old sign premises.

4 Fourteen exact intervals

Group the 64 LEFT support calls by their 13 radii. Within each group use the componentwise maximum of the recorded preceding-resource bounds r, b. The initial call is a separate row. The fixed table is:

R	r	b	Calls
5/2	603319/500000	$32128627/10^8$	17
101/40	475374/390625	8224681/25000000	5
51/20	61183419/50000000	$33662533/10^8$	4
103/40	30833983/25000000	8698269/25000000	5
13/5	30994887/25000000	$35540061/10^8$	4
21/8	24988347/20000000	$36644311/10^8$	5
53/20	$125900813/10^8$	$37372893/10^8$	4
107/40	63303431/50000000	1201521/3125000	4
27/10	63753427/50000000	489197/1250000	4
109/40	$128341449/10^8$	20069857/50000000	4
11/4	25857241/20000000	$41146991/10^8$	4
111/40	$129968327/10^8$	$42077471/10^8$	3
121/40	71458741/50000000	$46459359/10^8$	1
96/25	7361/4000	$641562334877/10^{12}$	1

The first 13 intervals are $[R, 39/10]$; the final interval is $[96/25, 21/5]$. All are closed and use a single cell. For fixed $u > 0$, $W > 0$,

$$\partial_r P = cW(3u^2/2 + .384/42) > 0, \qquad \partial_b P = 3cuW > 0.$$

Thus componentwise grouping preserves an upper bound even if the two maxima come from different calls. The first five sign polynomials do not depend on r, b. Each call is used sequentially only after its own actual-law resource bounds are acquired.

Proposition 4 (Exact interval certificate)**.** *For every table row, the seven polynomials*

$$A, \quad H_4, \quad J, \quad W, \quad G, \quad -Q, \quad -P$$

are strictly positive on its specified closed interval.

Proof. For $p(u)=\sum_{i=0}^{d} a_i u^i$ on $[l,h]$, write $u=l+(h-l)t$ and

$$q_j=(h-l)^j\sum_{i=j}^{d} a_i\binom{i}{j}l^{i-j},\qquad B_k=\sum_{j=0}^{k} q_j\frac{\binom{k}{j}}{\binom{d}{j}}.$$

The binomial theorem proves $p(l+(h-l)t)=\sum_{k=0}^{d} B_k\binom{d}{k}t^k(1-t)^{d-k}$. The basis is nonnegative and sums to1 on $[0,1]$. The degrees are $3,4,5,5,10,3,11$, giving 48 coefficients per row. The fixed rational checker computes all $14\cdot 48=672$ coefficients, and every coefficient exceeds $1/4$. It also independently reconstructs

$$q_j=\binom{d}{j}\sum_{k=0}^{j}(-1)^{j-k}\binom{j}{k}B_k$$

for every polynomial. Every coefficient is printed in the acceptance log, so the finite arithmetic is reproducible without numerical optimization. This proves continuous-domain positivity, including both endpoints. □

5 Completing the rays

The retained direct Gaussian derivative estimate, valid for every raw H,t, is

$$|\partial_t H_n(H,t)|\le\tfrac{2}{5}(t^2/2+1/2+e_n),\qquad e_n=n\big(\sqrt{n/(n-1)}-1\big)-1/2<.384/n.$$

For completeness, set $a=48/125$ and $x=1/n$. Expansion gives

$$(1-x)(1+x/2+ax^2)^2-1=x^2\{2a-3/4-(a+1/4)x+(a^2-a)x^2-a^2x^3\}.$$

The braces decrease on $[0,1/42]$ and equal $118707/42875000>0$ at its right endpoint. Taking square roots proves the stated uniform scale correction.

For each group set

$$L(u)=\tfrac{2}{5}(u^2/2+1/2+.384/42)+cQ'(u).$$

At its splice U, the checker verifies all six strict inequalities $Q,Q',Q'',L,L',L''<0$. Since $Q'''=-6$ and $L''=2/5-6c<0$, these signs persist for all $u\ge U$. For $\eta=\pm1$ the contact envelope $H_n(H,\eta u)+cQ(u)$ has derivative at most $L(u)<0$. Its value at U is already negative by Proposition 4. Hence both full exterior rays are excluded. For each individual call the group caps majorize Q and Q'. This conclusion applies to atoms, arbitrarily small masses, and every support point, with no limiting law or bounded-threshold substitution.

Combining the closed interval with the complete rays yields the original strict support conclusion $|X|<R$ at every actual $n\ge 42$, conditional on the retained contact/heat analysis and the actual preceding moment bounds.

6 Dependency savings and unresolved work

The new upper-index composition needs 19 of the old 31 finite native actions and all 152 LOW row producers. Six positive-ceiling/support actions and six special 43 bootstrap actions are bypassed by Lemma 2 and sharp predecessors. If the full uniform LOCAL42 transfer is accepted, the separate 55-action tail package is also unnecessary in this new upper-index module. Thus its conservative count is $238\to 171$, before new composition joins. This is a task count, not measured runtime or a repository-wide saving.

The grouped support certificate replaces only the support polynomial subchecks; it does not remove the remaining resource, confinement, profile, TINY/MID/LOW or harmonic obligations. A separate source audit found repeated LOW evaluations by producers and full binder replay,

but no one-pass receipt implementation is included. All 152 producers remain in the conservative plan.

A genuine BASE41 remains an input. The current mathematical floor 40 route states a sharp 38/39-to40/41/42 induction without the old upper-index tail. The complete pinned current provider body was unavailable in the inspected sources, while an older global 40 inventory explicitly consumes tail 43 and positive ceilings. Consequently neither executable base closure nor repository-wide obsolescence of those computations is established. Attainment at equality, published moment inequalities, analytic contact/Fourier/normalization licenses and strict Bernoulli comparison also remain separate premises.

7 Reproduction and preserved reference

The fixed checker is `mathematics/simplification/2026-10-27/check_grouped_support.py`. It imports only a new source-binding helper, never archived programs. The first acceptance returned 0 in about 0.12 seconds, with nine rejection controls and all exact coefficients stored under `runs/grouped-support-20261027/accept01/`. Other bounded checks, independent derivations, source locations and limitations are in the adjacent Markdown reports and `reviews/` directory.

The immutable original baseline is commit `6346fc1b4432bc5982822aa696eb566ed9e1e1ee`. The backend archive SHA256 begins `43b606138f826b47e`; full pins are in the reports and receipts. No inherited certificate, theorem statement, acceptance target or production runner is changed. No native heavy certificate program was run. This document proves new conditional implications and a small exact support certificate, not the full conjecture.